Spark Data Sink API

DataFrameWriter API : https://spark.apache.org/docs/latest/api/python/pyspark.sql.html#pyspark.sql.DataFrameWriter

General structure

```
DataFrameWriter
    .format(...)
    .option(...)
    .partitionBy(...)
    .bucketBy(...)
    .sortBy(...)
    .save()
```

Indicative Example

```
DataFrame.write
    .format("parquet")
    .mode(saveMode)
    .option("path", "/data/flights/")
    .save()
```

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Spark Data Sink API

DataFrameWriter API : https://spark.apache.org/docs/latest/api/python/pyspark.sql.html#pyspark.sql.DataFrameWriter

General structure

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    .format(...)
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    .partitionBy(...)
    .bucketBy(...)
    .sortBy(...)
    .save()
```

Indicative Example

```
DataFrame.write
        .format("parquet")
        .mode(saveMode)
        .option("path", "/data/flights/")
        .save()
```

Read Mode

1. append
2. overwrite
3. errorIfExists
4. ignore

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Spark File Layout

1. Number of files and file size
2. Partitions and Buckets
3. Sorted data

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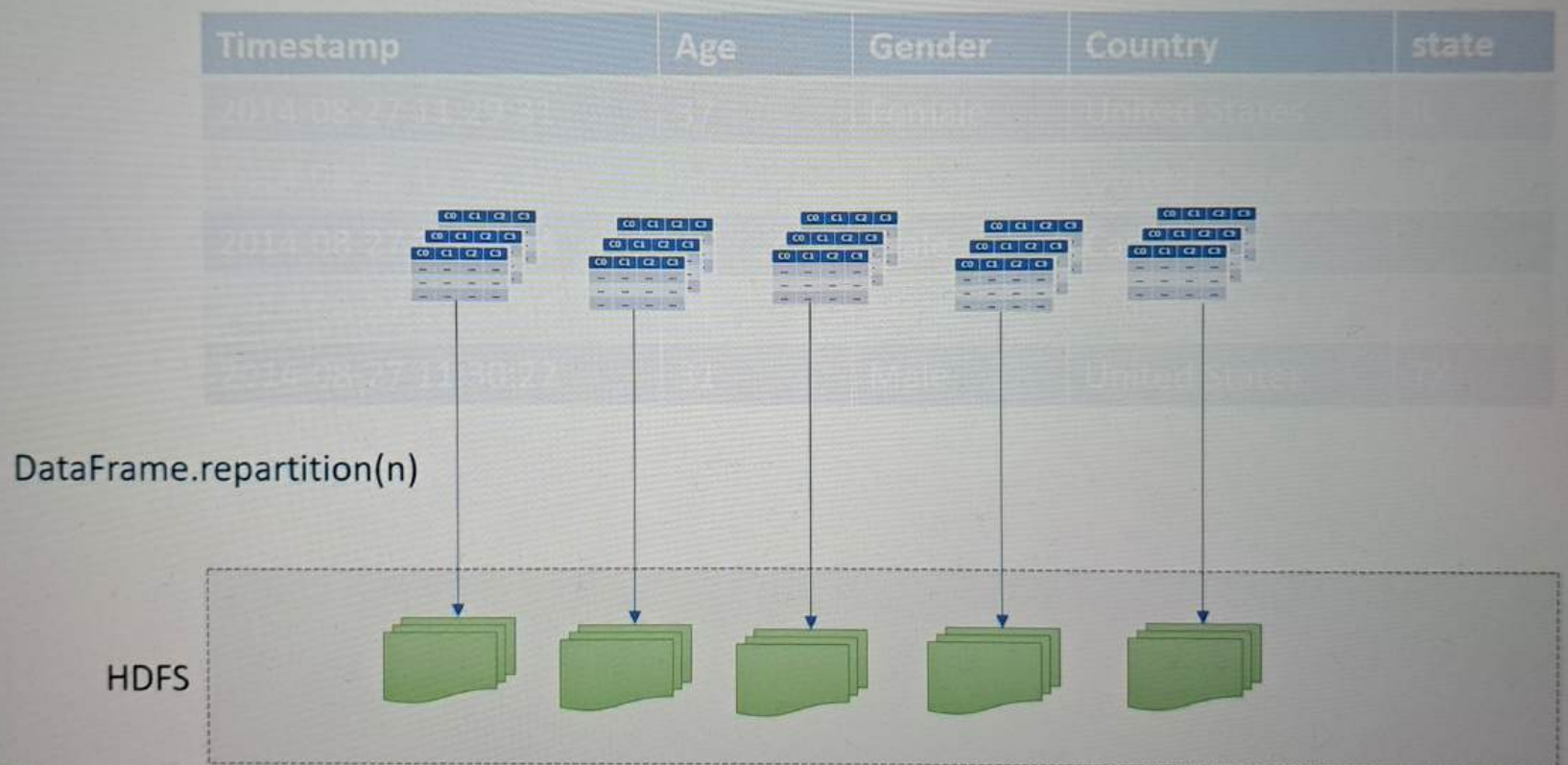
Spark Data Sink API



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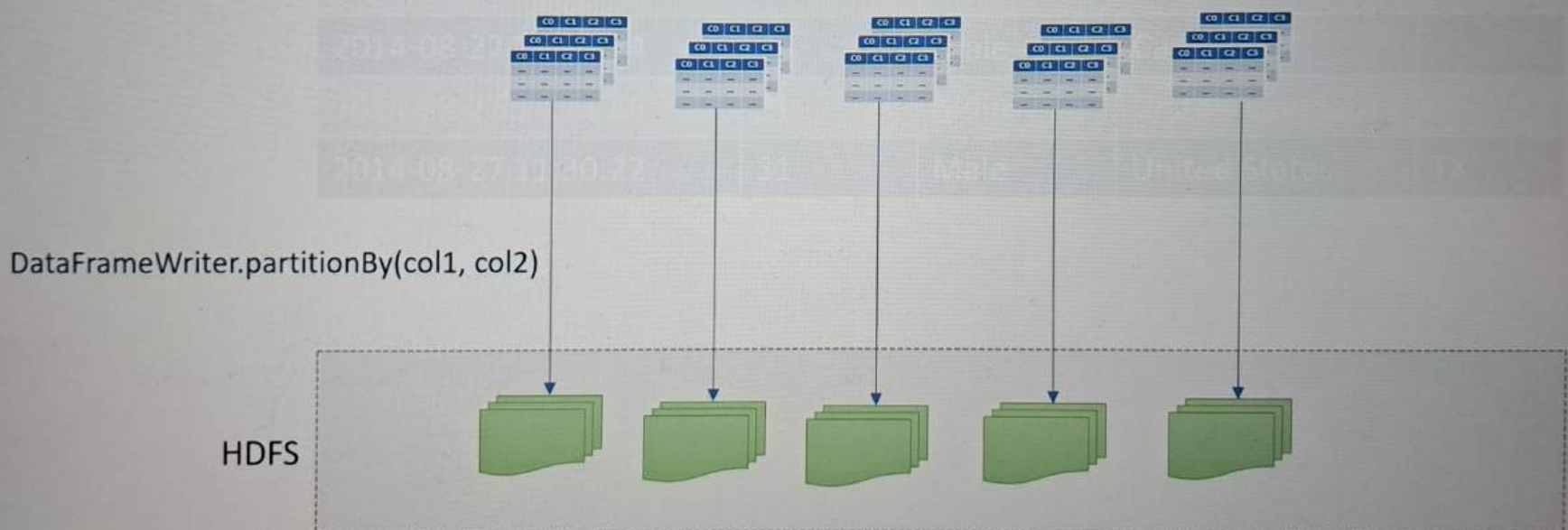
Spark Data Sink API

Timestamp	Age	Gender	Country	state
2014-08-27 11:29:34	37	Female	United States	IL
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2014-08-27 [illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2014-08-27 11:30:22	31	Male	United States	TX



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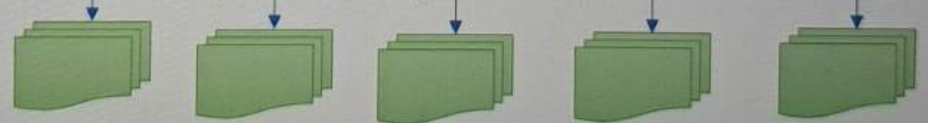
Spark Data Sink API

Timestamp	Age	Gender	Country	state
2014-08-27 11:29:03	37	Female	United States	IL
2014-08-27 11:29:08	35	Male	United States	TX
2014-08-27 11:29:10	31	Male	United States	NY

DataFrameWriter.bucketBy(n, col1, col2)

HDFS



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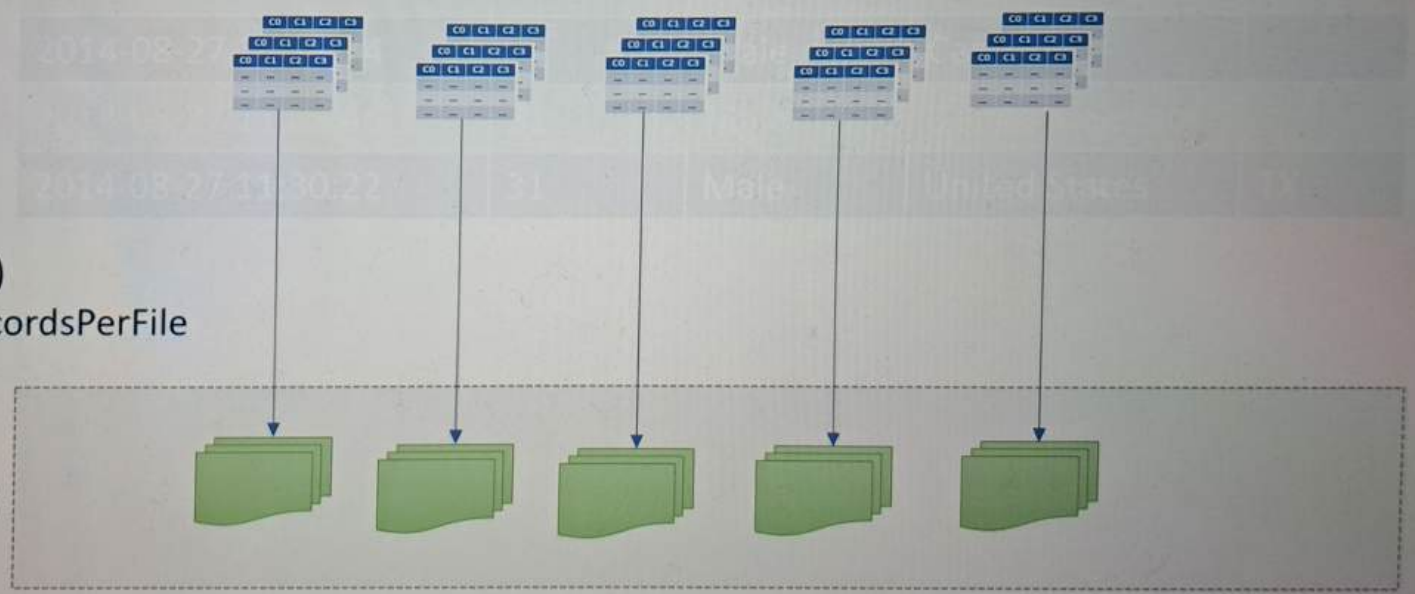
Spark Data Sink API

Timestamp	Age	Gender	Country	state
2014-08-27 11:29:31	37	Female	United States	IL
2014-08-27 [illegible]	28	[illegible]	[illegible]	[illegible]
2014-08-27 [illegible]	[illegible]	Male	[illegible]	[illegible]
2014-08-27 41:30:22	31	Male	United States	[illegible]

1. sortBy()
2. maxRecordsPerFile



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DataSinkDemo.py

```python
import ...

if __name__ == "__main__":
    spark = SparkSession \
        .builder \
        .master("local[3]") \
        .appName("SparkSchemaDemo") \
        .getOrCreate()

    logger = Log4j(spark)

    flightTimeParquetDF = spark.read \
        .format("parquet") \
        .load("dataSource/flight*.parquet")

    flightTimeParquetDF.write \
        .format("avro") \
        .mode("overwrite") \
        .option("path", "dataSink/avro/") \
        .save()
```

```
logger = Log4j(spark)

flightTimeParquetDF = spark.read \
    .format("parquet") \
    .load("dataSource/flight*.parquet")

flightTimeParquetDF.write \
    .format("avro") \
    .mode("overwrite") \
    .option("path", "dataSink/avro/") \
    .save()

logger.info("Num Partitions before: " + flightTimeParquetDF.rdd.getNumPartitions())
```

File Edit View Navigate Code Refactor Run Tools VCS Window Help 05-DataSinkDemo - DataSinkDemo.py - PyCharm

05-DataSinkDemo > DataSinkDemo.py

Project

- 05-DataSinkDemo C:\Users\prashant\Documen
  - app-logs
  - dataSink
    - avro
      - _SUCCESS.crc
      - .part-00000-be95c890-ebb1-4c46-880
      - _SUCCESS
      - part-00000-be95c890-ebb1-4c46-880
  - dataSource
  - lib
  - DataSinkDemo.py
  - log4j.properties
- External Libraries
- Scratches and Consoles

DataSinkDemo.py

```
flightTimeParquetDF = spark.read \
    .format("parquet") \
    .load("dataSource/flight*.parquet")

flightTimeParquetDF.write \
    .format("avro") \
    .mode("overwrite") \
    .option("path", "dataSink/avro/") \
    .save()

logger.info("Num Partitions before: " + str(flightTimeParquetDF.rdd.getNumPartitions()))
flightTimeParquetDF.groupBy(spark_partition_id()).count().show()
```

Run: DataSinkDemo

```
    confs: [default]
    0 artifacts copied, 2 already retrieved (0kB/16ms)
20/06/15 12:45:59 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
20/06/15 12:46:13 INFO SparkSchemaDemo: Num Partitions before: 2
+--------------------+------+
|SPARK_PARTITION_ID()| count|
+--------------------+------+
|                   0|470477|
+--------------------+------+

Process finished with exit code 0
```

9: Git 6: TODO 4: Run Terminal Python Console Event Log

PEP 8: W391 blank line at end of file

27:2 CRLF UTF-8 4 spaces Python 3.7 (pyspark) master

File Edit View Navigate Code Refactor Run Tools VCS Window Help 05-DataSinkDemo - DataSinkDemo.py - PyCharm

05-DataSinkDemo / DataSinkDemo.py

Project

- 05-DataSinkDemo C:\Users\prashant\Documen
  - app-logs
  - dataSink
    - avro
      - _SUCCESS.crc
      - .part-00000-be95c890-ebb1-4c46-880
      - _SUCCESS
      - part-00000-be95c890-ebb1-4c46-880
  - dataSource
  - lib
  - DataSinkDemo.py
  - log4j.properties
- External Libraries
- Scratches and Consoles

DataSinkDemo.py

```
logger.info("Num Partitions before: " + str(flightTimeParquetDF.rdd.getNumPartitions()))
flightTimeParquetDF.groupBy(spark_partition_id()).count().show()

partitionedDF = flightTimeParquetDF.repartition(5)
logger.info("Num Partitions after: " + str(partitionedDF.rdd.getNumPartitions()))
partitionedDF.groupBy(spark_partition_id()).count().show()

partitionedDF.write \
    .format("avro") \
    .mode("overwrite") \
    .option("path", "dataSink/avro/") \
    .save()
```

Run: DataSinkDemo

```
20/06/15 12:51:51 INFO SparkSchemaDemo: Num Partitions after: 5
+--------------------+------+
|SPARK_PARTITION_ID()|count|
+--------------------+------+
|                   1|94096|
|                   3|94095|
|                   4|94095|
|                   2|94096|
|                   0|94095|
+--------------------+------+

[Stage 26:==============================>                    (1 + 1) / 2]
```

9: Git 6: TODO 4: Run Terminal Python Console Event Log

PEP 8: W391 blank line at end of file

31:4 CRLF UTF-8 4 spaces Python 3.7 (pyspark) master

File Edit View Navigate Code Refactor Run Tools VCS Window Help 05-DataSinkDemo - DataSinkDemo.py - PyCharm

05-DataSinkDemo > DataSinkDemo.py

Project

05-DataSinkDemo C:\Users\prashant\Documents\GitHub\Spark-Programming-In-Pytho

- app-logs
- dataSink
  - avro
    - _SUCCESS.crc
    - .part-00000-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro.crc
    - .part-00001-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro.crc
    - .part-00002-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro.crc
    - .part-00003-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro.crc
    - .part-00004-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro.crc
    - _SUCCESS
    - part-00000-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro
    - part-00001-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro
    - part-00002-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro
    - part-00003-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro
    - part-00004-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro
- dataSource
- lib

DataSinkDemo.py

```
logger.info("Num Partitions before: " + str(flightTimeParquetDF.rdd.getNumPartitions()))
flightTimeParquetDF.groupBy(spark_partition_id()).count().show()

partitionedDF = flightTimeParquetDF.repartition(5)
logger.info("Num Partitions after: " + str(partitionedDF.rdd.getNumPartitions()))
partitionedDF.groupBy(spark_partition_id()).count().show()

partitionedDF.write \
    .format("avro") \
    .mode("overwrite") \
    .option("path", "dataSink/avro/") \
    .save()
```

Run: DataSinkDemo

```
20/06/15 12:51:51 INFO SparkSchemaDemo: Num Partitions after: 5

+-------------------+-----+
|SPARK_PARTITION_ID()|count|
+-------------------+-----+
|                  1|94096|
|                  3|94095|
|                  4|94095|
|                  2|94096|
|                  0|94095|
+-------------------+-----+

Process finished with exit code 0
```

1. Parallel Processing
2. Partition Elimination

9: Git 6: TODO 4: Run Terminal Python Console Event Log

PEP 8: W391 blank line at end of file 31:4 CRLF UTF-8 4 spaces Python 3.7 (pyspark) master

File Edit View Navigate Code Refactor Run Tools VCS Window Help 05-DataSinkDemo - DataSinkDemo.py - PyCharm

05-DataSinkDemo > DataSinkDemo.py

Project

05-DataSinkDemo C:\Users\prashant\Documents\GitHub\Spark-Programming-In-Pytho

- app-logs
- dataSink
  - avro
    - _SUCCESS.crc
    - .part-00000-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro.crc
    - .part-00001-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro.crc
    - .part-00002-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro.crc
    - .part-00003-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro.crc
    - .part-00004-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro.crc
    - _SUCCESS
    - part-00000-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro
    - part-00001-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro
    - part-00002-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro
    - part-00003-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro
    - part-00004-f0301d1f-ab66-4e89-b06e-8f31630c6be2-c000.avro
- dataSource
- lib

DataSinkDemo.py

```
19  logger.info("Num Partitions before: " + str(flightTimeParquetDF.rdd.getNumPartitions()))
20  flightTimeParquetDF.groupBy(spark_partition_id()).count().show()
21
22  partitionedDF = flightTimeParquetDF.repartition(5)
23  logger.info("Num Partitions after: " + str(partitionedDF.rdd.getNumPartitions()))
24  partitionedDF.groupBy(spark_partition_id()).count().show()
25
26  partitionedDF.write \
27      .format("avro") \
28      .mode("overwrite") \
29      .option("path", "dataSink/avro/") \
30      .save()
31
```

Run: DataSinkDemo

```
20/06/15 12:51:51 INFO SparkSchemaDemo: Num Partitions after: 5

+-------------------+-----+
|SPARK_PARTITION_ID()|count|
+-------------------+-----+
|                  1|94096|
|                  3|94095|
|                  4|94095|
|                  2|94096|
|                  0|94095|
+-------------------+-----+

Process finished with exit code 0
```

1. Parallel Processing
2. Partition Elimination

9: Git 6: TODO 4: Run Terminal Python Console Event Log

PEP 8: W391 blank line at end of file

31:4 CRLF UTF-8 4 spaces Python 3.7 (pyspark) master

```
'''
partitionedDF.write \
    .format("avro") \
    .mode("overwrite") \
    .option("path", "dataSink/avro/") \
    .save()
'''
flightTimeParquetDF.write \
    .format("json") \
    .mode("overwrite") \
    .option("path", "dataSink/json/") \
    .partitionBy("OP_CARRIER", "ORIGIN") \
    .save()
```

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05-DataSinkDemo 〉 dataSink

Project

- 05-DataSinkDemo C:\Users\prashant\Docu
  - app-logs
  - dataSink
    - avro
    - json
      - OP_CARRIER=AA
      - OP_CARRIER=AS
      - OP_CARRIER=CO
      - OP_CARRIER=DL
      - OP_CARRIER=HP
      - OP_CARRIER=NW
      - OP_CARRIER=TW
      - OP_CARRIER=UA
      - OP_CARRIER=US
      - OP_CARRIER=WN
      - ._SUCCESS.crc
      - _SUCCESS
  - dataSource
  - lib

DataSinkDemo.py

```
logger = Log4j(spark)

flightTimeParquetDF = spark.read \
        .format("parquet") \
        .load("dataSource/flight*.parquet")

logger.info("Num Partitions before: " + str(flightTimeParquet
flightTimeParquetDF.groupBy(spark_partition_id()).count().sho

partitionedDF = flightTimeParquetDF.repartition(5)
logger.info("Num Partitions after: " + str(partitionedDF.rdd.
partitionedDF.groupBy(spark_partition_id()).count().show()

'''
partitionedDF.write \
    .format("avro") \
    .mode("overwrite") \
    .option("path", "dataSink/avro/") \
```

File Edit View Navigate Code Refactor Run Tools VCS Window Help 05-DataSinkDemo - DataSinkDemo.py - PyCharm

05-DataSinkDemo 〉 dataSink

Project

- 05-DataSinkDemo C:\Users\prashant\Documents\GitHub\Spark-Prog
  - app-logs
  - dataSink
    - avro
    - json
      - OP_CARRIER=AA
        - ORIGIN=ABQ
        - ORIGIN=ALB
        - ORIGIN=AMA
        - ORIGIN=ATL
        - ORIGIN=AUS
        - ORIGIN=BDL
        - ORIGIN=BHM
        - ORIGIN=BNA
        - ORIGIN=BOS
        - ORIGIN=BUF
        - ORIGIN=BUR
        - ORIGIN=BWI
        - ORIGIN=CLE

DataSinkDemo.py

```
13      logger = Log4j(spark)
14
15      flightTimeParquetDF = spark.read \
16          .format("parquet") \
17          .load("dataSource/flight*.parquet")
18
19      logger.info("Num Partitions before: " + str(
20      flightTimeParquetDF.groupBy(spark_partition_
21
22      partitionedDF = flightTimeParquetDF.repartit
23      logger.info("Num Partitions after: " + str(p
24      partitionedDF.groupBy(spark_partition_id()).
25
26      '''
27      partitionedDF.write \
28          .format("avro") \
29          .mode("overwrite") \
30          .option("path", "dataSink/avro/") \
```

File Edit View Navigate Code Refactor Run Tools VCS Window Help 05-DataSinkDemo - DataSinkDemo.py - PyCharm

05-DataSinkDemo 〉 dataSink

Project

- 05-DataSinkDemo C:\Users\prashant\Documents\GitHub\Spark-Programming-In-Python
  - app-logs
  - dataSink
    - avro
    - json
      - OP_CARRIER=AA
        - ORIGIN=ABQ
        - ORIGIN=ALB
        - ORIGIN=AMA
        - ORIGIN=ATL
        - ORIGIN=AUS
        - ORIGIN=BDL
        - ORIGIN=BHM
        - ORIGIN=BNA
        - ORIGIN=BOS
          - .part-00000-1cce2bd7-fd60-4b80-9c71-9451b900445e.c000.json.crc
          - part-00000-1cce2bd7-fd60-4b80-9c71-9451b900445e.c000.json
        - ORIGIN=BUF
        - ORIGIN=BUR
        - ORIGIN=BWI
        - ORIGIN=CLE
        - ORIGIN=CLT
        - ORIGIN=CMH

DataSinkDemo.py

```
logger = Log4j(spark)

flightTimeParquetDF = spark.read \
    .format("parquet") \
    .load("dataSource/flight*.parquet")

logger.info("Num Partitions before: " + str(flightT
flightTimeParquetDF.groupBy(spark_partition_id()).c

partitionedDF = flightTimeParquetDF.repartition(5)
logger.info("Num Partitions after: " + str(partitio
partitionedDF.groupBy(spark_partition_id()).count()

'''
partitionedDF.write \
    .format("avro") \
    .mode("overwrite") \
    .option("path", "dataSink/avro/") \
    .save()
'''
flightTimeParquetDF.write \
    .format("json") \
```

File Edit View Navigate Code Refactor Run Tools VCS Window Help 05-DataSinkDemo - part-00000-1cce2bd7-fd60-4b80-9c71-9451b900445e.c000.json - PyCharm

05-DataSinkDemo > dataSink > json > OP_CARRIER=AA > ORIGIN=BOS > part-00000-1cce2bd7-fd60-4b80-9c71-9451b900445e...

DataSinkDemo

Project

- 05-DataSinkDemo C:\Users\prashant\Documents\GitHub\Spark-Programming-In-Python
  - app-logs
  - dataSink
    - avro
    - json
      - OP_CARRIER=AA
        - ORIGIN=ABQ
        - ORIGIN=ALB
        - ORIGIN=AMA
        - ORIGIN=ATL
        - ORIGIN=AUS
        - ORIGIN=BDL
        - ORIGIN=BHM
        - ORIGIN=BNA
        - ORIGIN=BOS
          - .part-00000-1cce2bd7-fd60-4b80-9c71-9451b900445e.c000.json.crc
          - part-00000-1cce2bd7-fd60-4b80-9c71-9451b900445e.c000.json
        - ORIGIN=BUF
        - ORIGIN=BUR
        - ORIGIN=BWI
        - ORIGIN=CLE
        - ORIGIN=CLT
        - ORIGIN=CMH
        - ORIGIN=COS
        - ORIGIN=CVG
        - ORIGIN=DAL
        - ORIGIN=DAY
        - ORIGIN=DCA
        - ORIGIN=DEN
        - ORIGIN=DFW
        - ORIGIN=DRO
        - ORIGIN=DSM
        - ORIGIN=DTW
        - ORIGIN=EGE
        - ORIGIN=ELP

DataSinkDemo.py | part-00000-1cce2bd7-fd60-4b80-9c71-9451b900445e.c000.json


```
1409 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":851,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"ORD","DEST_CITY_NAME":"Ch
1410 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":1043,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"ORD","DEST_CITY_NAME":"C
1411 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":1103,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"ORD","DEST_CITY_NAME":"C
1412 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":1177,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"ORD","DEST_CITY_NAME":"C
1413 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":1200,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"ORD","DEST_CITY_NAME":"C
1414 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":1353,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"ORD","DEST_CITY_NAME":"C
1415 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":1365,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"ORD","DEST_CITY_NAME":"C
1416 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":1433,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"ORD","DEST_CITY_NAME":"C
1417 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":1527,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"ORD","DEST_CITY_NAME":"C
1418 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":1793,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"ORD","DEST_CITY_NAME":"C
1419 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":1797,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"ORD","DEST_CITY_NAME":"C
1420 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":1931,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"ORD","DEST_CITY_NAME":"C
1421 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":2353,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"ORD","DEST_CITY_NAME":"C
1422 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":225,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"SAN","DEST_CITY_NAME":"Sa
1423 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":401,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"SJU","DEST_CITY_NAME":"Sa
1424 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":1019,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"SJU","DEST_CITY_NAME":"S
1425 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":1971,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"SJU","DEST_CITY_NAME":"S
1426 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":1983,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"FLL","DEST_CITY_NAME":"F
1427 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":11,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"LAX","DEST_CITY_NAME":"Los
1428 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":163,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"LAX","DEST_CITY_NAME":"Lo
1429 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":181,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"LAX","DEST_CITY_NAME":"Lo
1430 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":223,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"LAX","DEST_CITY_NAME":"Lo
1431 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":189,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"SEA","DEST_CITY_NAME":"Se
1432 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":109,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"SJC","DEST_CITY_NAME":"Sa
1433 {"FL_DATE":"2000-01-31","OP_CARRIER_FL_NUM":129,"ORIGIN_CITY_NAME":"Boston, MA","DEST":"SJC","DEST_CITY_NAME":"Sa
1434
```


9: Git 6: TODO 4: Run Terminal Python Console Event Log

1:1 LF UTF-8 2 spaces No JSON schema Python 3.7 (pyspark) master

File Edit View Navigate Code Refactor Run Tools VCS Window Help 05-DataSinkDemo - DataSinkDemo.py - PyCharm

05-DataSinkDemo > dataSink

Project

05-DataSinkDemo C:\Users\prashant\Documents\GitHub\Spark-Programming-In-Python
- app-logs
- dataSink
- dataSource
- lib
- DataSinkDemo.py
- log4j.properties
- External Libraries
- Scratches and Consoles

DataSinkDemo.py

```python
logger = Log4j(spark)

flightTimeParquetDF = spark.read \
    .format("parquet") \
    .load("dataSource/flight*.parquet")

logger.info("Num Partitions before: " + str(flightTimeParquetDF.rdd.getNumPartitions()))
flightTimeParquetDF.groupBy(spark_partition_id()).count().show()

partitionedDF = flightTimeParquetDF.repartition(5)
logger.info("Num Partitions after: " + str(partitionedDF.rdd.getNumPartitions()))
partitionedDF.groupBy(spark_partition_id()).count().show()

'''
partitionedDF.write \
    .format("avro") \
    .mode("overwrite") \
    .option("path", "dataSink/avro/") \
    .save()
'''
flightTimeParquetDF.write \
    .format("json") \
    .mode("overwrite") \
    .option("path", "dataSink/json/") \
    .partitionBy("OP_CARRIER", "ORIGIN") \
    .option("maxRecordsPerFile", 10000) \
    .save()
```

Git TODO Run Terminal Python Console

Event Log

40:1 CRLF UTF-8 4 spaces Python 3.7 (pyspark) master

File Edit View Navigate Code Refactor Run Tools VCS Window Help 05-DataSinkDemo - DataSinkDemo.py - PyCharm

05-DataSinkDemo 〉 dataSink

Project

- 05-DataSinkDemo C:\Users\prashant\Documents\GitHub\Spark-Programming-In-Python
  - app-logs
  - dataSink
    - avro
    - json
      - OP_CARRIER=AA
      - OP_CARRIER=AS
      - OP_CARRIER=CO
      - OP_CARRIER=DL
        - ORIGIN=ABE
        - ORIGIN=ABQ
        - ORIGIN=AGS
        - ORIGIN=ALB
        - ORIGIN=ANC
        - ORIGIN=ATL
          - .part-00000-47c5ed04-e3a9-47c8-a6c6-bfd0c71841f9.c000.json.crc
          - .part-00000-47c5ed04-e3a9-47c8-a6c6-bfd0c71841f9.c001.json.crc
          - part-00000-47c5ed04-e3a9-47c8-a6c6-bfd0c71841f9.c000.json
          - part-00000-47c5ed04-e3a9-47c8-a6c6-bfd0c71841f9.c001.json
        - ORIGIN=AUS
        - ORIGIN=BDL
        - ORIGIN=BHM
        - ORIGIN=BIL
        - ORIGIN=BNA

DataSinkDemo.py

```
14
15      flightTimeParquetDF = spark.read \
16          .format("parquet") \
17          .load("dataSource/flight*.parquet")
18
19      logger.info("Num Partitions before: " + str(flightTime
20      flightTimeParquetDF.groupBy(spark_partition_id()).cou
21
22      partitionedDF = flightTimeParquetDF.repartition(5)
23      logger.info("Num Partitions after: " + str(partitione
24      partitionedDF.groupBy(spark_partition_id()).count().s
25
26      '''
27      partitionedDF.write \
28          .format("avro") \
29          .mode("overwrite") \
30          .option("path", "dataSink/avro/") \
31          .save()
32      '''
33      flightTimeParquetDF.write \
34          .format("json") \
35          .mode("overwrite") \
36          .option("path", "dataSink/json/") \
```

Spark Data Sink API

DataFrameWriter API : https://spark.apache.org/docs/latest/api/python/pyspark.sql.html#pyspark.sql.DataFrameWriter

General structure

```
DataFrameWriter
    .format(...)
    .option(...)
    .partitionBy(...)
    .bucketBy(...)
    .sortBy(...)
    .save()
```

1. How to use DataFrameWriter API
2. How to use PartitionBy(...)
3. → How to control file size using maxRecordsPerFile

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